

SQL-BASED ANALYSIS OF AN E-COMMERCE SALES DATASET USING MICROSOFT SQL SERVER

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Introduction

Data analysis plays an important role in helping organisations understand business performance, customer behaviour, and sales trends. This project presents an exploratory analysis of an e-commerce sales dataset using Microsoft SQL Server. The objective was to extract meaningful business insights from transactional data by analysing customer activity, product performance, revenue generation, marketing effectiveness, and payment preferences.

The analysis was conducted using SQL queries to answer key business questions and identify patterns that could support data-driven decision-making.

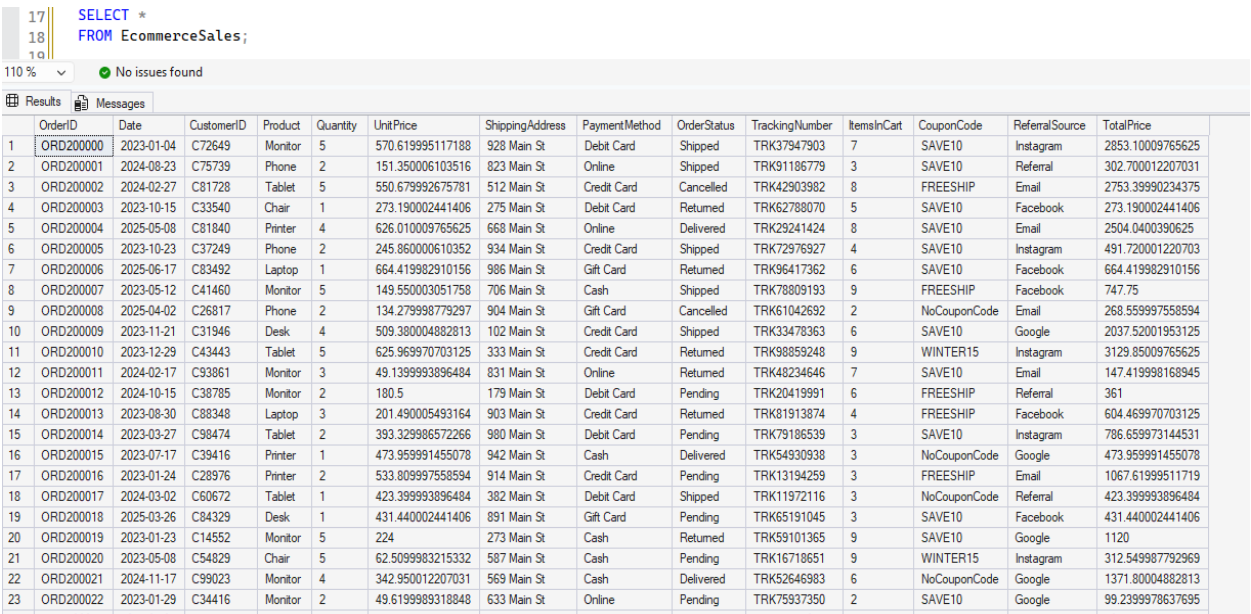


Figure 1: Structure of the e-commerce sales dataset imported into SQL Server.

Dataset Description

The dataset contains transactional records from an e-commerce business covering the period from January 2023 to June 2025. The dataset includes information relating to customers, products, order dates, quantities purchased, payment methods, coupon codes, referral sources, and transaction values.

The available variables provided sufficient information to perform customer, product, marketing, and revenue analyses.

Objectives

The objectives of this project were to:

- Analyse customer purchasing behaviour.

- Identify top-performing products.
- Evaluate coupon effectiveness.
- Assess marketing channel performance.
- Analyse payment method preferences.
- Investigate revenue trends over time.
- Generate business insights using SQL queries.

Methodology

The dataset was imported into Microsoft SQL Server and analysed using Structured Query Language (SQL). Various SQL techniques were applied, including filtering, aggregation, grouping, sorting, and date-based analysis.

The analysis was organised into the following categories:

- Dataset Overview
- Product Analysis
- Marketing Analysis
- Payment Analysis
- Revenue Analysis
- Customer Behaviour Analysis

Dataset Overview

The initial stage of the analysis focused on understanding the size and characteristics of the dataset. SQL queries were used to determine the total number of orders, customer records, unique customers, and repeat customers.

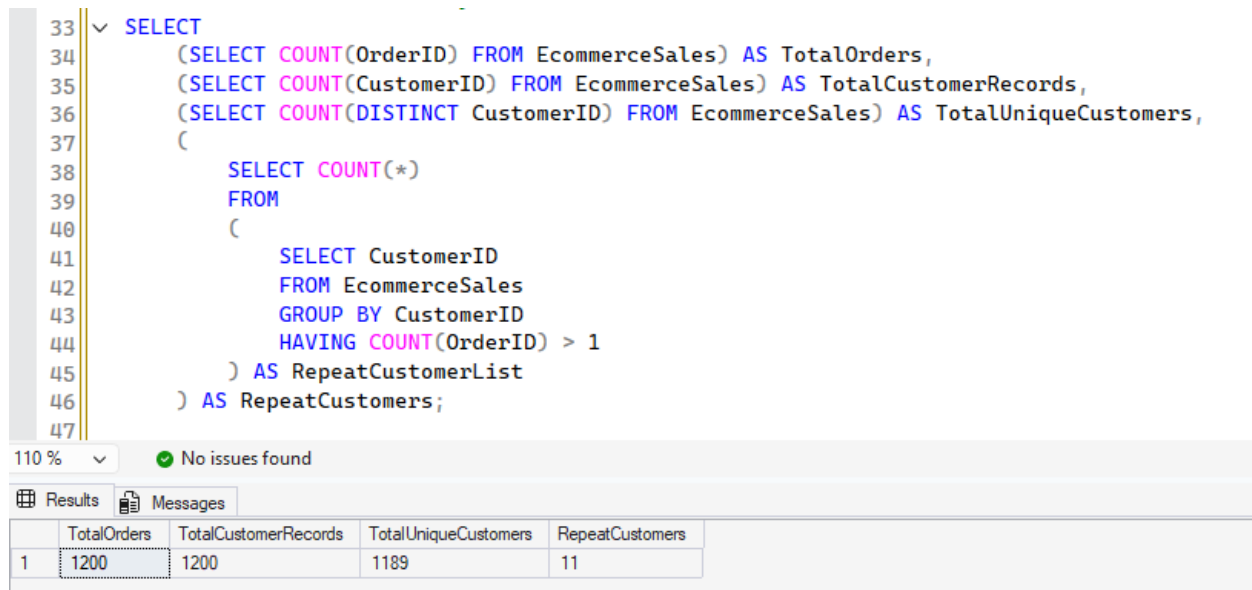


Figure 2: Dataset Overview Summary

The analysis provided a general understanding of customer activity and established a foundation for subsequent investigations. The dataset contained 1,200 transaction records and 1,189 unique customers, indicating that repeat purchasing behaviour was relatively uncommon within the observed period.

Product Analysis

Product performance was evaluated by analysing both revenue generation and sales volume across the product catalogue. Revenue analysis helped identify the products contributing most to overall business performance, while units sold provided insight into customer purchasing preferences and product demand.

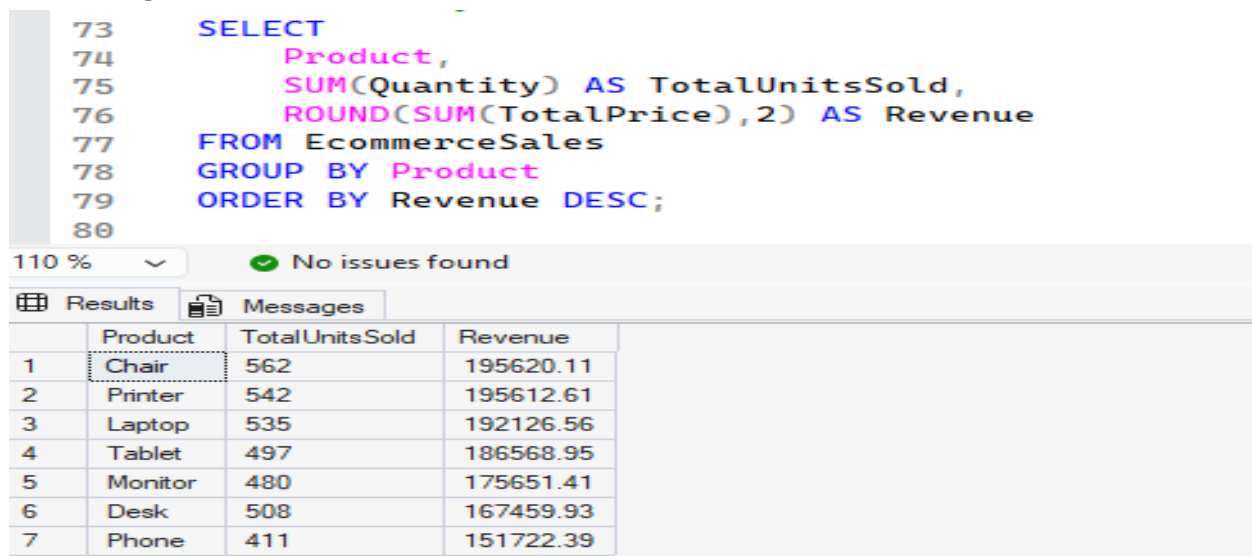


Figure 3: Product revenue and units sold analysis.

The analysis revealed noticeable differences in product performance across the catalogue. Chairs generated the highest revenue (£195,620.11), closely followed by Printers (£195,612.61) and Laptops (£192,126.56). These products also recorded relatively high sales volumes, indicating strong customer demand. In contrast, Phones generated the lowest revenue (£151,722.39) and sold the fewest units, suggesting comparatively weaker market performance. Overall, the findings indicate that a relatively small group of products accounted for a substantial proportion of total business revenue, highlighting their importance to overall sales performance.

Marketing Analysis

Marketing effectiveness was examined by analysing referral source performance and coupon usage. Referral sources were evaluated based on their contribution to order volume and revenue generation, while coupon analysis was used to assess the effectiveness of promotional campaigns in influencing customer purchasing behaviour and spending patterns.

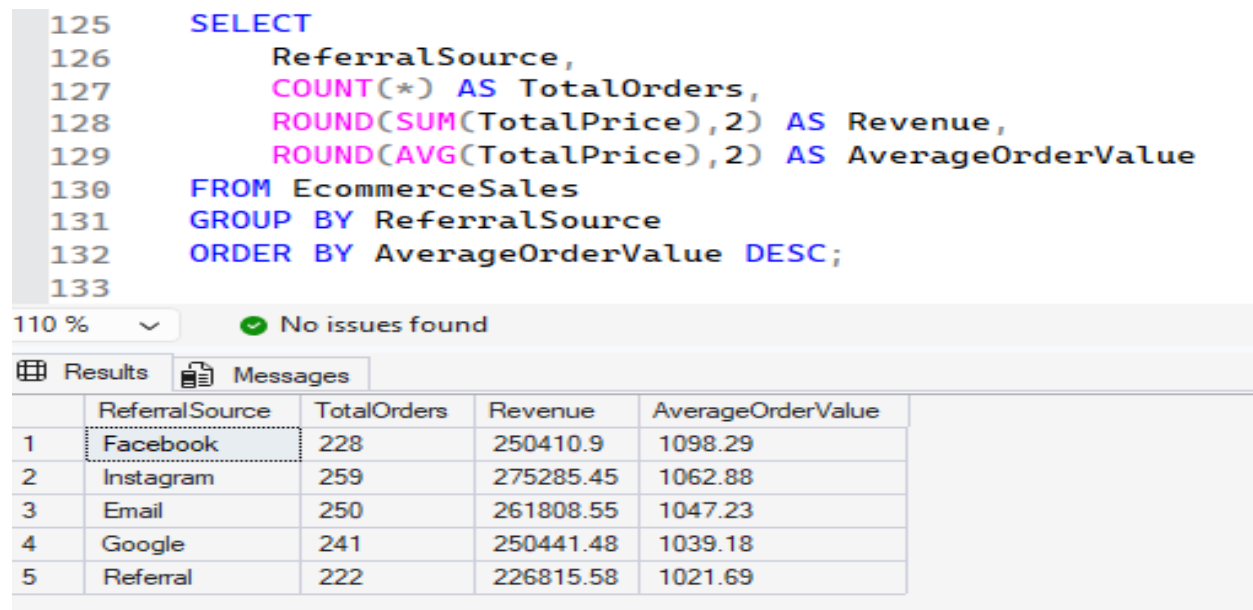


Figure 5: Referral source performance by orders and revenue.

The analysis revealed notable differences in the effectiveness of referral channels. Instagram generated the highest revenue (£275,285.45) and the largest number of orders (259), making it the strongest overall marketing channel. However, Facebook recorded the highest average order value (£1,098.29), indicating that customers acquired through Facebook tended to spend more per transaction. Referral generated the lowest revenue (£226,815.58) and the lowest average order value (£1,021.69), suggesting comparatively weaker performance. These findings indicate that different marketing channels contribute differently to both customer acquisition and customer spending behaviour.

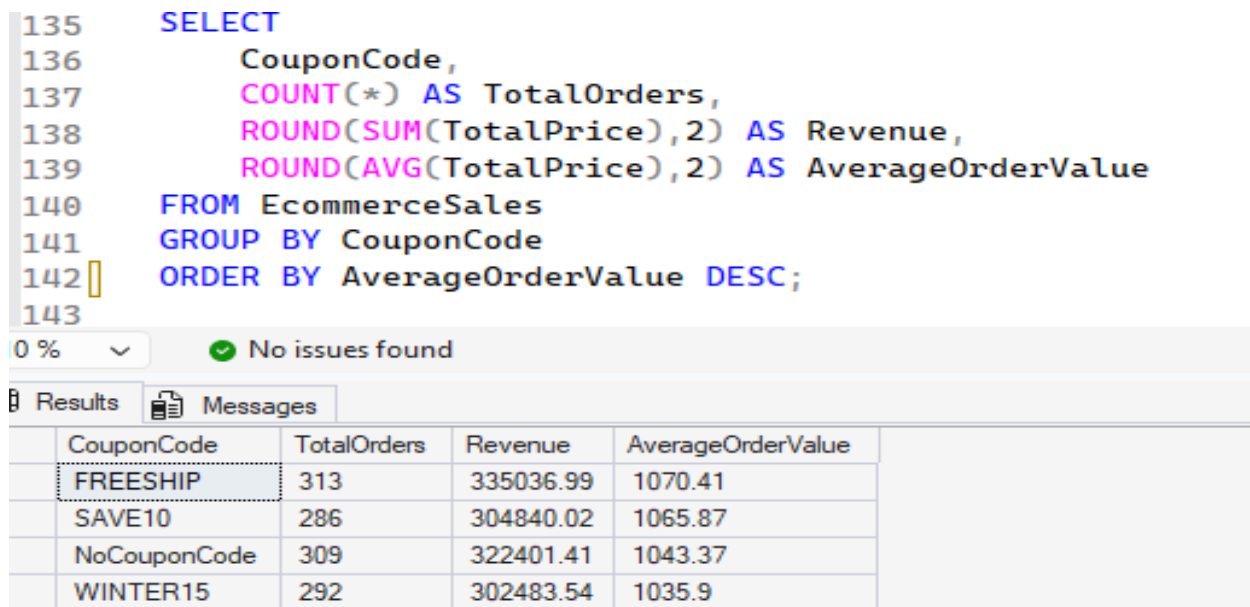


Figure 6: Coupon performance analysis.

Coupon analysis revealed differences in revenue generation, order volume, and average order value across promotional campaigns. The FREESHIP campaign generated the highest revenue (£335,036.99), the highest number of orders (313), and the highest average order value (£1,070.41), making it the most effective promotional strategy within the dataset. SAVE10 also performed strongly, generating £304,840.02 in revenue with an average order value of £1,065.87. Customers who did not use a coupon generated substantial revenue (£322,401.41), indicating that promotional discounts were not the sole driver of purchasing activity. WINTER15 produced the lowest revenue (£302,483.54) and the lowest average order value (£1,035.90), suggesting comparatively weaker performance. Overall, the findings indicate that shipping-related incentives were more effective at encouraging customer spending than percentage-based discount promotions.

Payment Analysis

Customer payment preferences were investigated by analysing payment method usage, revenue contribution, and average order values. This analysis aimed to identify the most frequently used payment methods and determine whether certain payment options were associated with higher customer spending levels.

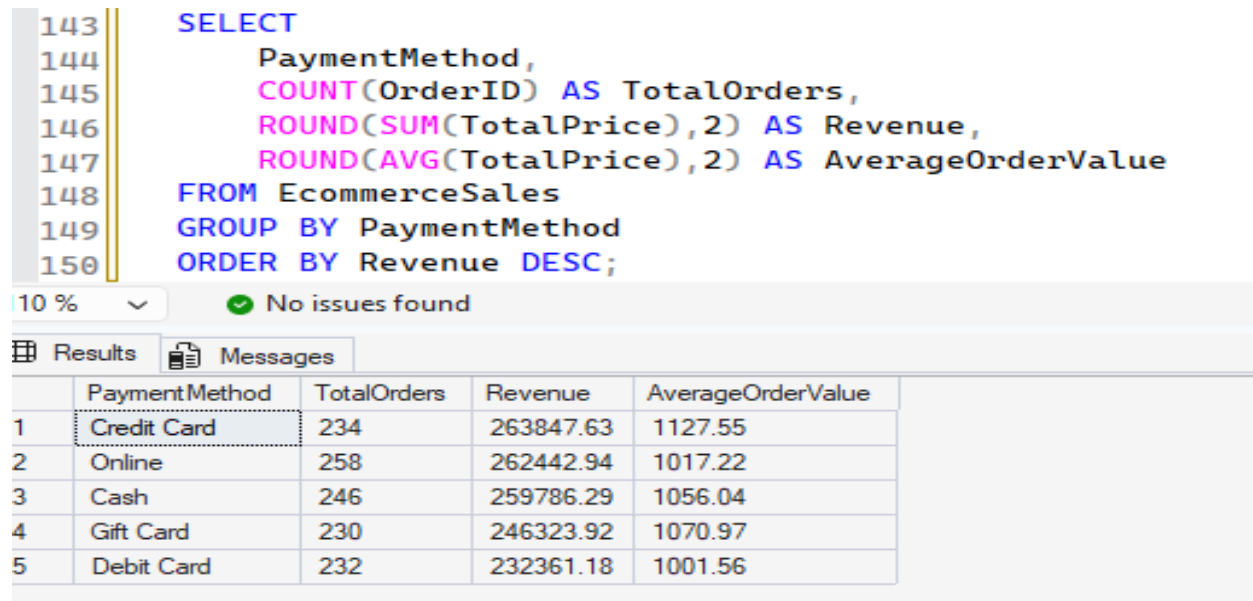


Figure 7: Payment method analysis.

The analysis showed that customers used all payment methods at relatively similar rates, with total orders ranging from 230 to 258. Online payments recorded the highest number of orders (258), while Gift Card transactions recorded the fewest (230). Despite having fewer orders than Online payments, Credit Card transactions generated the highest revenue (£263,847.63) and the highest average order value (£1,127.55), indicating that customers using credit cards tended to spend more per transaction. Debit Card transactions generated the lowest revenue (£232,361.18) and the lowest average order value (£1,001.56). Overall, the findings suggest that while payment method preferences were fairly balanced, customers using credit cards contributed greater value to the business through higher spending levels.

Order Status Analysis

Order fulfilment performance was examined by analysing the distribution of order statuses. This analysis provides insight into the progression of orders through different stages of the fulfilment process and highlights the relative frequency of each order outcome.

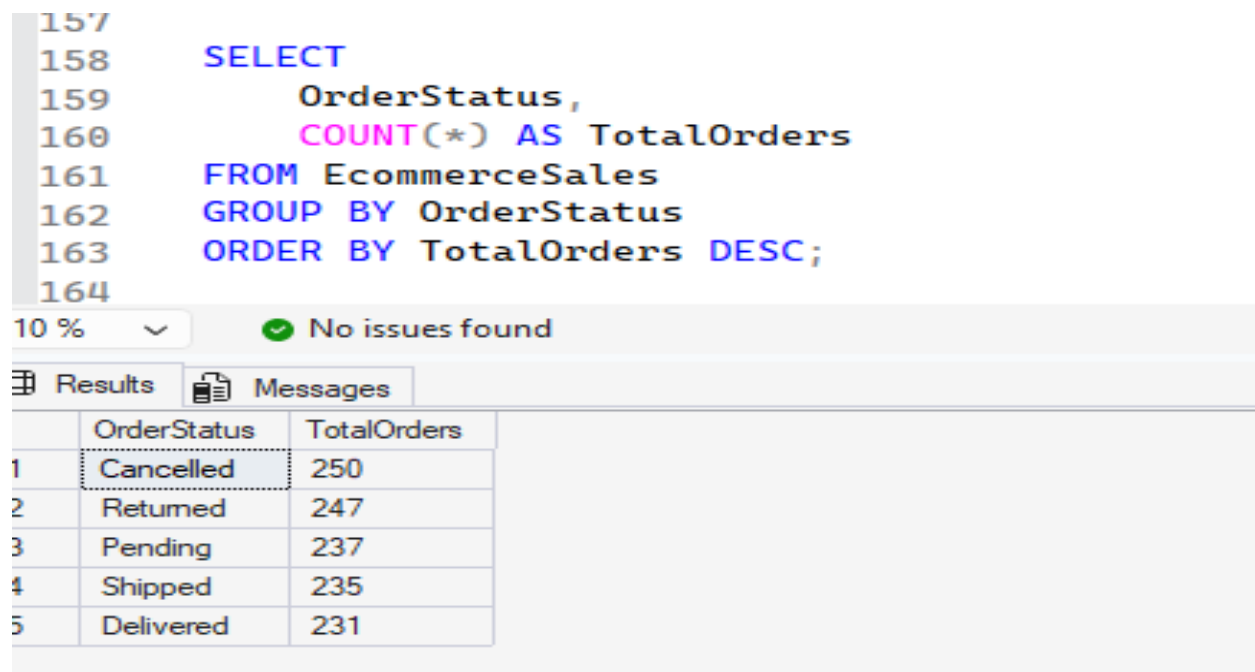


Figure 8: Distribution of orders by status.

The distribution of order statuses was relatively balanced across all categories, with Cancelled orders recording the highest count (250) and Delivered orders recording the lowest count (231). The differences between categories were minimal, indicating that no single order status dominated the dataset. While real-world e-commerce businesses would typically be expected to record a substantially higher proportion of delivered orders, the balanced distribution observed in this dataset provides a useful basis for analysing order status patterns and demonstrating SQL aggregation techniques.

Revenue Analysis

Revenue performance was examined using various descriptive metrics, including total revenue, average order value, and yearly revenue trends.

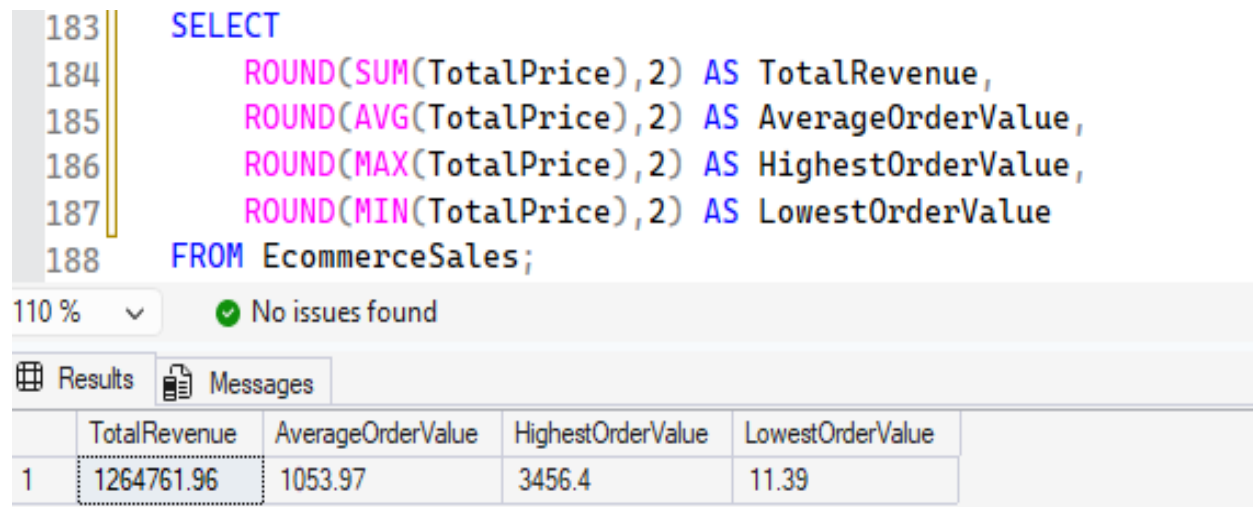


Figure 9: Revenue summary statistics.

The dataset generated a total transaction value of £1,264,761.96, with an average order value of £1,053.97. The highest transaction value was £3,456.40, while the lowest transaction value was £11.39.

Revenue Recognition Consideration

A comparison between total transaction value and revenue associated with shipped and delivered orders revealed a substantial difference. While the dataset contained £1,264,761.96 in total recorded transaction value, only £488,759.90 was associated with completed orders. This observation demonstrates the importance of considering order status when interpreting revenue metrics. In real-world business environments, cancelled and returned orders would generally be excluded from realised revenue calculations.

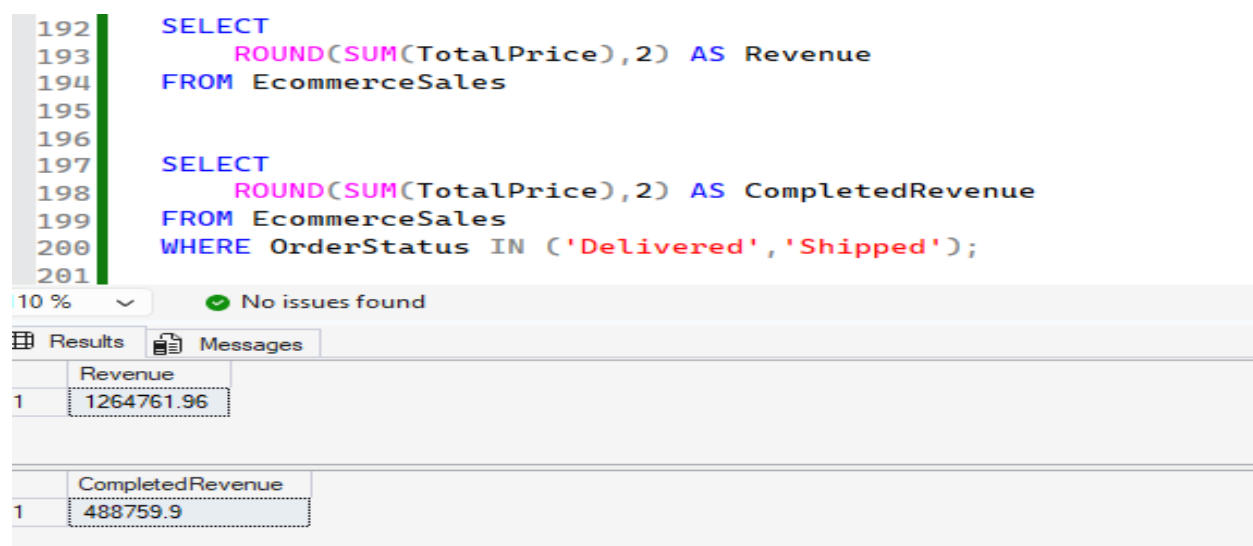


Figure 10: Comparison between total transaction value and completed-order revenue.

Revenue Trend Analysis

Initially, yearly revenue analysis suggested a significant decline in business performance.

```
221 SELECT
222     YEAR(Date) AS SalesYear,
223     COUNT(OrderID) AS TotalOrder,
224     ROUND(SUM(TotalPrice),2) AS YearlyRevenue
225 FROM EcommerceSales
226 GROUP BY YEAR(Date)
227 ORDER BY SalesYear;
228
```

110 % ✓ No issues found

Results Messages

	SalesYear	TotalOrder	YearlyRevenue
1	2023	510	552643.24
2	2024	459	480235.87
3	2025	231	231882.85

Figure 11: Revenue generated by year.

The initial analysis suggested a substantial decline in business performance over time. Revenue decreased from £552,643.24 in 2023 to £480,235.87 in 2024 and further declined to £231,882.85 in 2025. Similarly, total orders decreased from 510 in 2023 to 459 in 2024 and 231 in 2025. At first glance, these results appeared to indicate a significant deterioration in sales performance.

However, further investigation revealed that the 2025 data covered only the period up to June 2025. To ensure a fair comparison, equivalent January–June periods were analysed across all years.

```
240 SELECT
241     YEAR(Date) AS SalesYear,
242     COUNT(OrderID) AS TotalOrder,
243     ROUND(SUM(TotalPrice),2) AS Revenue
244 FROM EcommerceSales
245 WHERE MONTH(Date) <= 6
246 GROUP BY YEAR(Date)
247 ORDER BY SalesYear;
```

110 % ✓ No issues found

Results Messages

	SalesYear	TotalOrder	Revenue
1	2023	252	286501.52
2	2024	237	257059.34
3	2025	231	231882.85

Figure 12: January–June revenue comparison across years.

When equivalent January–June periods were compared across all years, the decline in revenue remained evident but was less severe than initially suggested. Revenue decreased

from £286,501.52 in 2023 to £257,059.34 in 2024 and further declined to £231,882.85 in 2025. Total orders also fell from 252 in 2023 to 237 in 2024 and 231 in 2025. These findings indicate a gradual reduction in business performance rather than the dramatic decline implied by the full-year comparison.

The results suggest that revenue reduction was associated with declining order volumes over time. To better understand the underlying causes of this trend, further analysis was conducted on customer activity and purchasing behaviour across the same period.

Customer Behaviour Analysis

Further analysis was conducted to determine the factors contributing to the decline in revenue.

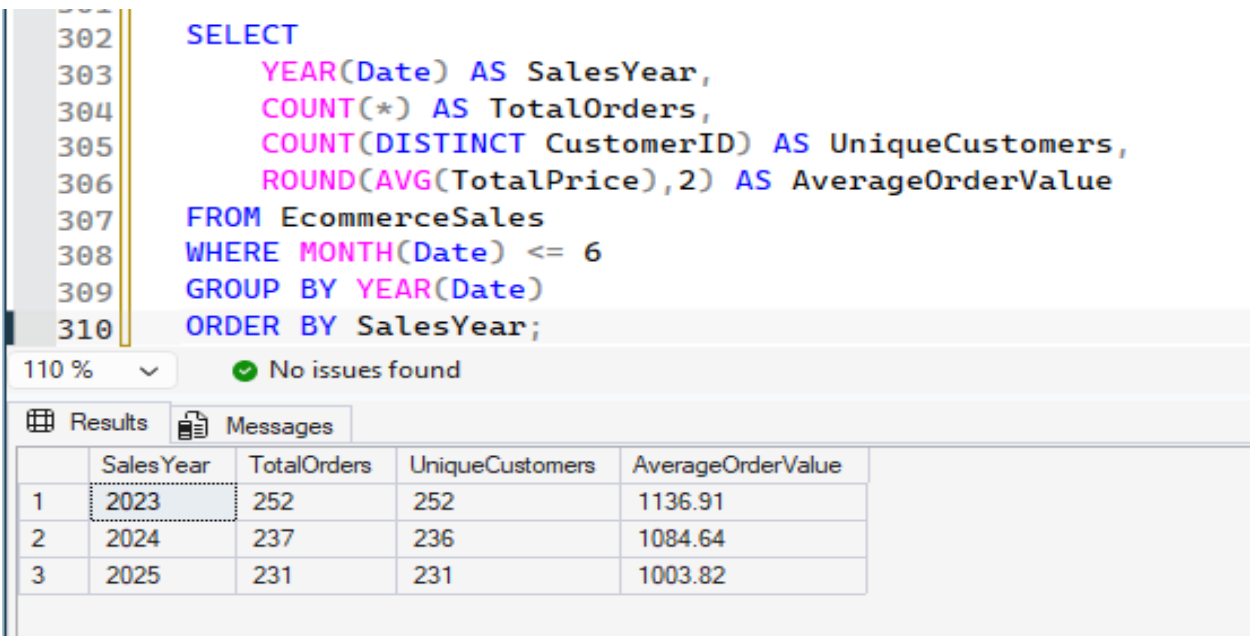


Figure 13: Orders, unique customers, and average order value comparison.

The analysis revealed that:

- Total orders declined from 252 in 2023 to 231 in 2025.
- Unique customers declined from 252 in 2023 to 231 in 2025.
- Average order value decreased from £1,136.91 in 2023 to £1,003.82 in 2025.

These findings indicate that the decline in revenue was driven by a combination of decreasing customer numbers, reduced order volumes, and lower spending per transaction.

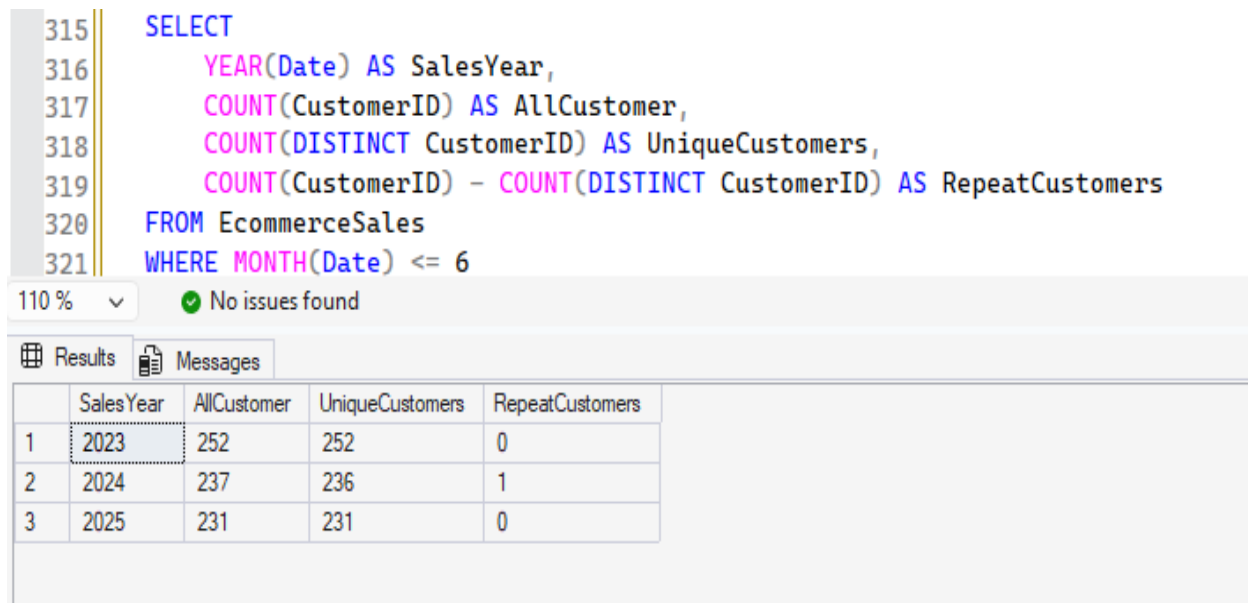


Figure 14: Repeat customer analysis.

The analysis revealed extremely low levels of repeat purchasing during the first six months of each year. In 2023 and 2025, the number of customer records was identical to the number of unique customers, indicating that no customers made multiple purchases. In 2024, only one customer returned to make an additional purchase. These findings suggest that revenue generation relied almost entirely on acquiring new customers rather than retaining existing ones.

Key Business Insights

The analysis generated several important business insights:

- The FREESHIP coupon campaign produced the highest revenue and average order value, indicating strong customer engagement with shipping-related promotions.
- Referral sources varied significantly in their contribution to revenue and customer spending, suggesting differences in marketing channel effectiveness.
- Product performance differed considerably across the product catalogue, with some products generating substantially higher revenue and sales volumes than others.
- A comparison of total transaction value and completed-order revenue revealed that a significant proportion of recorded transaction value was associated with cancelled, returned, or pending orders. This highlights the importance of considering order status when evaluating business performance.

- Revenue showed a gradual downward trend between 2023 and 2025. After adjusting for the partial-year nature of the 2025 data, the decline remained evident across comparable January–June periods.
- Customer numbers, order volumes, and average order values all declined during the analysis period, contributing to the reduction in overall revenue.
- Repeat purchasing behaviour was extremely limited, indicating that most customers made only a single purchase during the observed period. This suggests that the business relied heavily on customer acquisition rather than customer retention.

Conclusion

This project demonstrated the practical application of Microsoft SQL Server for business-oriented data analysis. Through customer, product, marketing, payment, order fulfilment, and revenue analyses, meaningful patterns were identified within the e-commerce dataset.

The analysis revealed that declining revenue performance was associated with reductions in customer numbers, order volumes, and average spending per transaction. The findings also highlighted several opportunities for improving business performance through more effective customer retention, marketing, and revenue management strategies.

The project further highlighted the importance of analysing completed-order revenue separately from total transaction value and ensuring fair comparisons across equivalent reporting periods when evaluating business performance.

Overall, the analysis demonstrates how SQL can transform raw transactional data into actionable business insights that support operational decision-making, marketing strategy, customer retention initiatives, and long-term business planning.